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Dedicated or unified GPU memory

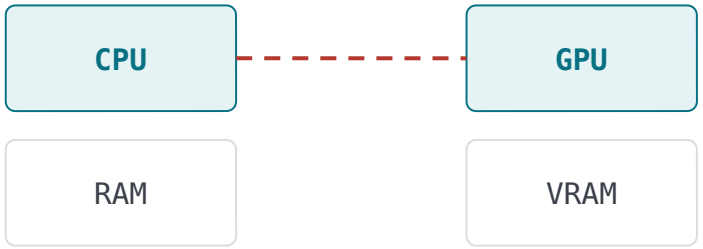
Setup and use cases, side by side. A technical module for people who run models on their own machines.

TECHNICAL · DRAFT FOR REVIEW

Brought by SYGNIF · reviewed before publication · AGPL-3.0 with the contributor's name

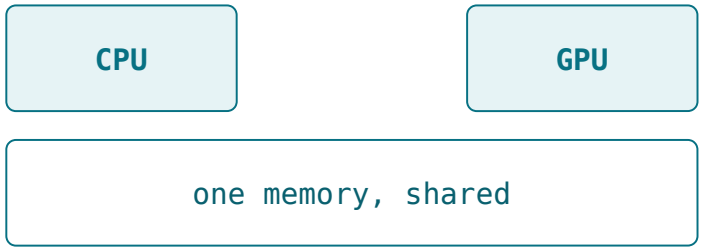
One pool, or two pools joined by a bus

Dedicated



PCIe: every byte the GPU needs from RAM is copied across

Unified



Both work on the same bytes; nothing is copied, the pool is the limit

Where you meet them

Dedicated

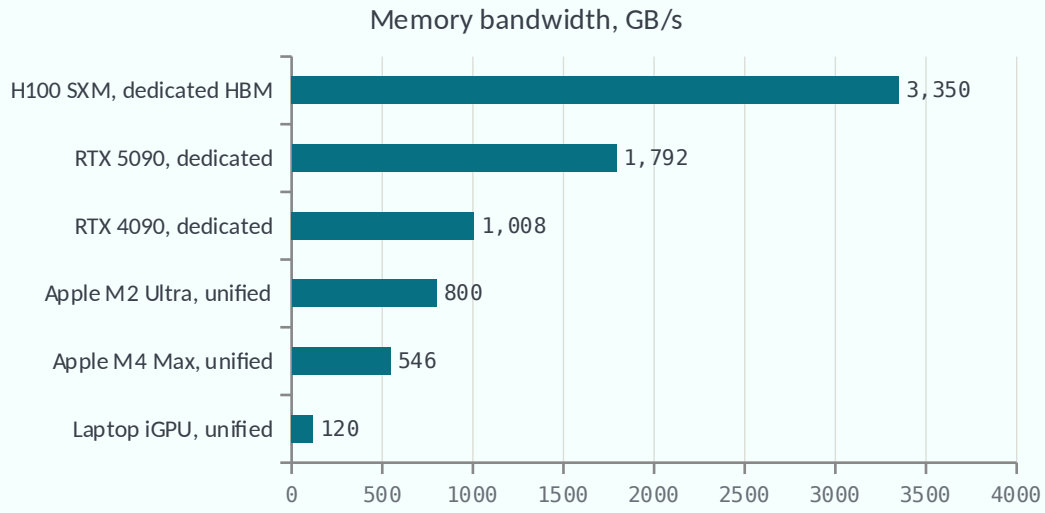
- Desktop and workstation cards: NVIDIA GeForce and RTX Pro, AMD Radeon
- Datacenter accelerators with HBM: NVIDIA H100/H200/B200, AMD Instinct MI300
- Rented by the hour: RunPod, Lambda, cloud GPU instances
- Several cards in one machine, joined by PCIe or NVLink

Unified

- Apple silicon: MacBook, Mac mini, Mac Studio (M-series)
- AMD Ryzen AI Max (Strix Halo) laptops and mini PCs
- Intel Core Ultra laptops with the integrated GPU
- NVIDIA DGX Spark (GB10) desktop, game consoles

Two numbers decide most cases

Capacity says whether a model fits at all. Bandwidth says how fast it runs once it fits: for generating text, the ceiling in tokens per second is bandwidth divided by the bytes read per token.



Capacity, typical

Dedicated
8–32 GB consumer cards · 48–96 GB workstation · 80–192 GB HBM datacenter

Unified
16–192 GB on Mac Studio · 128 GB on Strix Halo and DGX Spark · the OS decides how much the GPU may take

Setting up a dedicated GPU

1

Drivers and runtime

NVIDIA CUDA or AMD ROCm; versions of driver, runtime and framework must match.

2

Power and cooling

300 to 600 W per card; the power supply, the case and the room have to carry it.

3

The bus

PCIe 4.0 or 5.0 x16. Data that lives in RAM is copied before the GPU can use it.

4

More than one card

Models can be split across cards; NVLink or PCIe joins them. Complexity rises with the count.

5

Upgradable

A card is swapped; the machine stays. Memory grows by buying a bigger card.

6

Where it lives

A desktop, a rack, or rented by the hour. Not a laptop, not a quiet room.

Setting up unified memory

1

Nothing to install

The operating system brings the runtime: Metal and MLX on macOS, ROCm or Vulkan on AMD, DirectML on Windows.

2

Shared by rule

The OS decides how much of the pool the GPU may claim, often around three quarters; some systems let you raise it.

3

One device

No second GPU, no NVLink. What you buy is what you run.

4

Fixed at purchase

Memory is soldered. Decide the size on day one; it does not grow later.

5

Quiet and portable

Laptops and small desktops, tens of watts, no fan noise in a classroom.

6

Tooling

Ollama, llama.cpp, LM Studio and MLX run out of the box; CUDA-only software does not.

Where a dedicated GPU wins

Training and fine-tuning

Gradients want bandwidth; a run of hours on a card is a run of days on unified memory.

Serving many users

Throughput scales with bandwidth and with the number of cards.

Rendering, games, CAD

Frame rates and viewport speed are bandwidth-bound.

Models that fit in VRAM

If the model fits in the card's memory, the card is the fastest place for it.

Batch work

Embeddings for a whole library, transcribing a term of recordings: throughput jobs.

Growing needs

Add or swap cards; rent bigger ones by the hour when a project needs them.

Where unified memory wins

Big models, one person

A 70B model at 4-bit needs about 40 GB: it fits in 64 GB of unified memory, not in a 24 GB card.

Long context

The conversation memory (KV cache) grows with context length; a large pool absorbs it.

Development on a laptop

Build and test locally, then train or serve elsewhere.

Quiet rooms

Classrooms, offices, bedrooms: tens of watts, no fan.

Data that must stay

Private notes, pupils' work, patient records: never leave the machine.

No bus to feed

When the data set is larger than any VRAM, sharing beats copying.

Three questions, one answer each

1	**Does the model fit in one card's VRAM?**	Yes: dedicated is faster. No: unified memory, or several cards.
2	**Do you train, or only run?**	Train: dedicated. Run for yourself: either; unified if it must be quiet or portable.
3	**How many people at once?**	One: unified is enough. Many: dedicated, and more than one card.

Speed follows bandwidth, room follows capacity. Buy for the model you will actually run.

Five questions for a school or a small team

1. **Which models, how large?**
   Parameters and quantisation set the memory you need; add a third for context and system.

2. **How many users at the same time?**
   One person tolerates two tokens per second; a class of 25 does not.

3. **Train, fine-tune, or only run?**
   Training changes the answer to dedicated almost always.

4. **Where will it live?**
   Desk, rack, bag, classroom: noise, heat and power are part of the price.

5. **What does a year of power cost?**
   A 450 W card at four hours a day is about 650 kWh a year; a unified laptop uses a tenth.

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This is a draft for the Hub

- Read it against what it claims: does the rule of thumb hold for your machine?
- Send back the slide you had to change: sygnif@entrosana.com, or through the Hub's status page.
- Published under AGPL-3.0 with the contributor's name once admitted; the record of its making stays checkable.
- A German version and a one-page worksheet follow after the first review.

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